

NBA four-factors composite model

Technical report: team strength, game-level win probability, and validation

Summary

The model turns eight four-factor rates into a team-strength composite fitted to regular-season win%, then converts the pre-game composite gap between two teams (plus a back-to-back flag) into a home-win probability. Data come from Basketball-Reference (season tables 2015-2026, excluding the 2020 bubble year, and team game logs 2023-2026).

- **Team level:** out-of-sample error is about 3.5 wins per team-season (R^2 about 0.92). Net rating alone explains slightly more (R^2 about 0.93), so the composite is an interpretable decomposition, not a better rating.
- **Game level:** about 66-68% accuracy and 0.616 log loss (leave-one-season-out, 4,289 games), against 55% for home court alone. Calibration is close to exact by quintile.
- **What helped:** a back-to-back term (about 0.003 log loss). Exponential decay of game weights (half-life about 25 games) is roughly neutral to slightly positive.
- **What did not:** strength-of-schedule adjustment, short rolling windows, venue-split composites, rest days beyond back-to-back, 3-in-4 and 4-in-6 flags.
- **Not tested:** betting lines, injuries and lineups, travel. Do not treat this as a market edge without a closing-line comparison.

1. Model specification

Features. For each team, eight quantities, all oriented so that higher is better:

Side	Feature	Definition
Offense	eFG%	$(FG + 0.5 \times 3P) / FGA$
Offense	Turnovers (negated)	TOV per 100 possessions
Offense	Offensive rebounds	ORB per 100 possessions
Offense	Free-throw rate	FTA / FGA
Defense	Opponent eFG% (negated)	Opponent $(FG + 0.5 \times 3P) / FGA$
Defense	Turnovers forced	Opponent TOV per 100
Defense	Opponent ORB (negated)	Opponent ORB per 100
Defense	Opponent FT rate (negated)	Opponent FTA / FGA

Possessions per game average both sides: $FGA + 0.4 FTA - 1.07 \times ORB / (ORB + opp DRB) \times (FGA - FG) + TOV$.

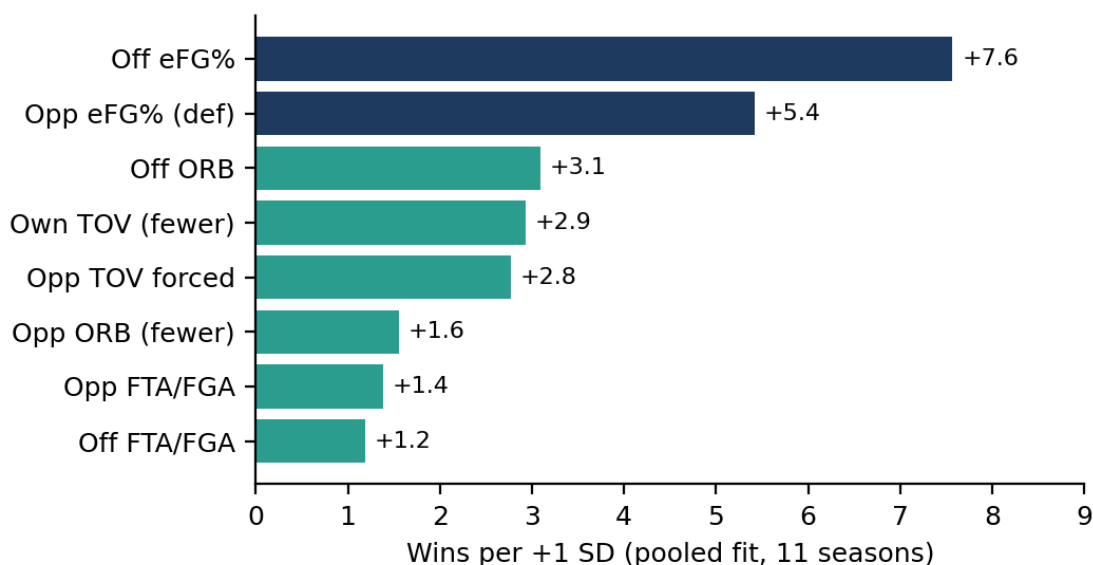
Composite. Features are centered on each season's league mean, divided by the pooled standard deviation, and combined with ridge-regression weights ($\lambda = 0.1$) fitted to win%. The composite is the weighted sum, expressed in win% points.

Game predictor. Before each game, each team's 16 raw totals (8 team, 8 opponent) over prior games are weighted by $0.5^{(\text{games ago} / 25)}$ and summed, then converted to rates. Never average per-game rates. With δ = home composite minus away composite (win% points) and $b2b_net$ = (away on a back-to-back) minus (home on a back-to-back):

$$P(\text{home win}) = 1 / (1 + \exp(-(a + b \times \delta + c \times b2b_net)))$$

Pooled fit on 2023-26: $a = 0.241$ (home edge, about 56% at equal teams), $b = 0.0374$ per win% point, $c = 0.327$. A gap of 10 win% points is worth about 10 points of win probability near even; a 20-point gap gives about 73%.

2. Composite weights



Shooting efficiency on both ends is roughly 46% of the composite. Offensive eFG% alone is worth 7.6 wins per standard deviation; free-throw rates are the least important terms. Weights come from the pooled fit of 11 seasons and are stable in sign across seasons; individual weights are noisy because the four factors are correlated and n is only 30 teams per season.

3. Team-season validation

Test	Result
Single-season fit, 2026 (in-sample)	R^2 0.964; leave-one-out RMSE 3.7 wins
2026 weights frozen, applied to 10 earlier seasons	Avg RMSE 3.8 wins; R^2 0.90 vs net rating 0.93; error rises from 2.6 (2025) to 4.7 (2015)
Pooled 11 seasons, leave-one-season-out	Avg RMSE 3.5 wins; R^2 0.915
Add strength of schedule (SOS)	No change: RMSE 3.48 and R^2 0.915 with and without. SOS coefficient -0.73 wins per point, but its SD is only 0.32

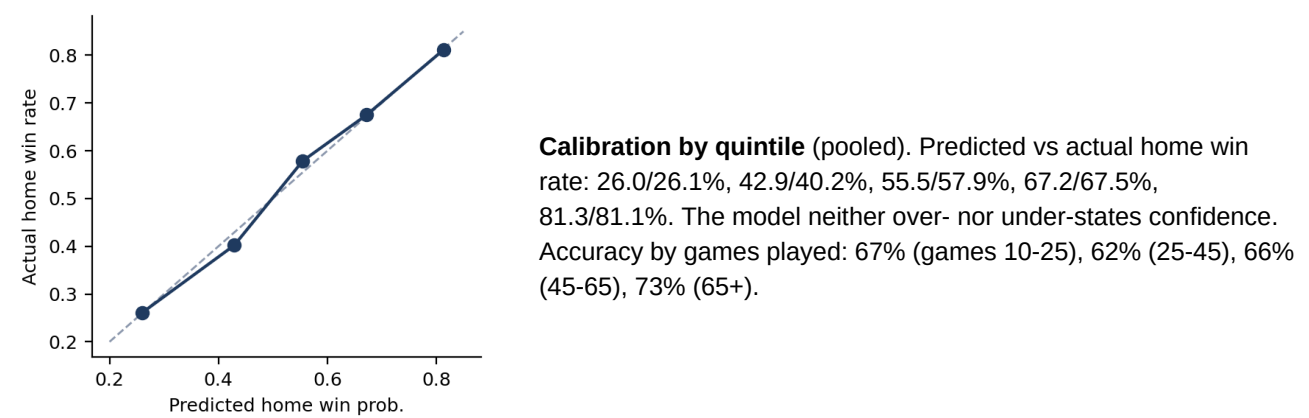
The frozen-weight decay in accuracy for older seasons reflects the three-point era: the value of each factor shifted. Pooling with within-season centering removes most of that drift. The NRTg comparison is generous to NRTg because its R^2 is refit in each season.

4. Game-level validation

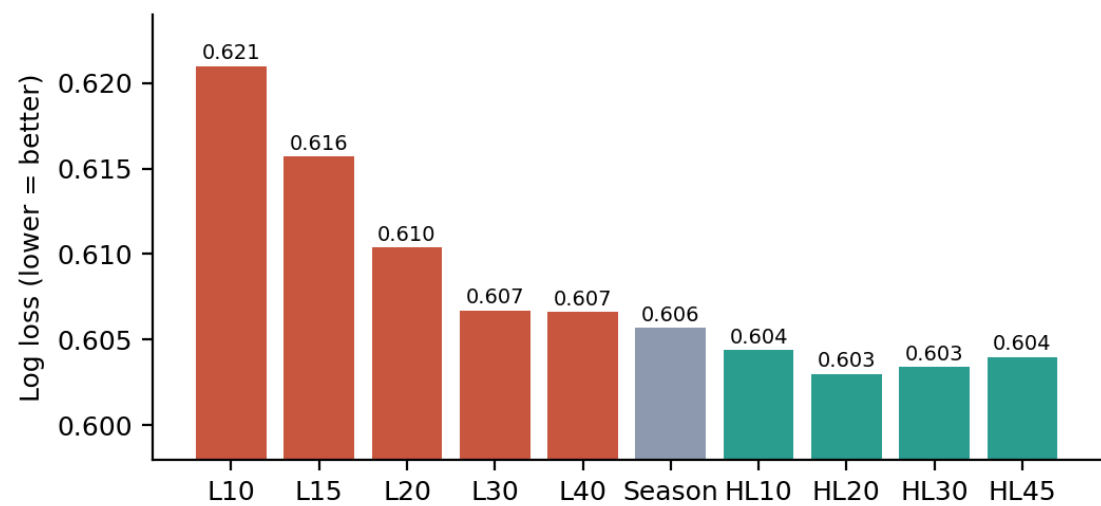
Two-season test (fit on one season, score the other, both directions, about 2,140 games):

Model	Accuracy	Log loss
Home court only	54.6%	0.689
Season-to-date margin (MOV)	67.4%	0.604
Composite (season-to-date)	68.0%	0.606
Composite + MOV	67.7%	0.604
Venue-split composite (home team at home vs away team on road)	66.5%	0.623

The composite correlates 0.98 with season-to-date margin per game, so the two are statistically indistinguishable; the differences (0.6 points of accuracy, 0.002 log loss) are inside sampling noise of about 1.4 points. Splitting by venue halves the sample per team and hurts.



5. Rolling windows and decay



Hard windows lose to season-to-date, monotonically worse as the window shrinks (last 10 games: 0.621 log loss vs 0.606 for the full season). Exponentially decayed weights with a half-life of 20-45 games score marginally better (0.603) than season-to-date; the curve is flat across that range. In the later four-season test the decay gain shrank to about 0.001, so the honest reading is "no cost, possibly a slight benefit." The half-life of 25 was chosen by comparing values on the same two seasons that are reported and should be confirmed on other data.

6. Rest and back-to-back games

Leave-one-season-out over 2023-26 (4,289 games; composite weights trained on 2015-22 only):

Model	Accuracy	Log loss	vs delta only
Decayed composite + back-to-back	65.9%	0.6156	-0.0026 (se 0.0012)
Decayed composite (delta only)	66.1%	0.6182	-
Season-to-date composite	65.7%	0.6193	+0.0011
Home court only	55.1%	-	-

- Relative to the composite, home teams on a back-to-back against a rested opponent won 8.5 points less often than predicted (n = 477); away teams on a back-to-back lost 5.7 points more often (n = 592); games with both tired showed about zero (n = 223).
- Coefficient c = 0.327 with standard error 0.067 (logit units). The effect is well determined but small in log-loss terms, and it does not improve accuracy.
- Rest days beyond back-to-back, 3-in-4 and 4-in-6 flags improved log loss by at most 0.0009 and were left out.

7. Limitations

- **No market benchmark.** Accuracy near 66-68% is typical of public power ratings. Compare log loss or closing-line value before any betting use; the market already prices back-to-backs.
- **No injuries, lineups, or travel.** Whole-team rates cannot see who is playing. Back-to-back effects interact with resting stars.
- **Few seasons at game level.** Only four seasons of game logs were tested. Window and half-life choices are within noise of one another.
- **Correlated features.** Individual weights are unstable; only the eFG% terms are reliably large.
- **Scraping.** Data are pulled from Basketball-Reference under rate limits; table layouts can change, which would break parsers.

8. Implementation and next steps

The reference implementation is `nba_composite.py` (download, fit-weights, backtest, fit-logit, score).

Recommended run order: download seasons and logs; fit-weights on completed seasons excluding the test seasons; backtest; fit-logit; then score each day with `--refresh` so logs include yesterday's results. Refit weights and the logit each preseason.

- Add closing lines and evaluate against the market (log loss and CLV).
- Add player availability (injury reports, minutes shares) as the main missing information.
- Pool more seasons of game logs to settle the half-life and test travel and altitude.
- Test a blend with last season's composite for the first 10-20 games of a season.